

Practical 6: Rendering External JSON Data using Fetch API, Search and Filter

Problem definition

Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

Key questions

1. How is JSON fetched, parsed, and rendered?

JSON data is fetched from an external JSON file using the Fetch API.

The fetched response is parsed into JavaScript objects or arrays.

The parsed data is then processed using JavaScript.

Finally, the data is dynamically rendered and displayed on the HTML page.

2. Which array methods are used for search, filter, sort, and pagination?

The `filter()` method is used to search and filter the required data from an array.

The `includes()` method is used to match the search text with the available data.

The `sort()` method is used to arrange the data in ascending or descending order.

The `slice()` method is used to divide the data into smaller portions for pagination.

3. How are loading and error states handled?

A loading state is displayed while the JSON data is being fetched. After the data is successfully loaded, the loading message is replaced with the actual data. If an error occurs while fetching or processing the data, an error-handling mechanism displays an appropriate error message to the user.

4. How is modularity maintained in JavaScript files?

Modularity is maintained by dividing the JavaScript code into separate functions or modules. Each function or module performs a specific task such as fetching, searching, filtering, sorting, or rendering data. This makes the code more organized, reusable, readable, and easier to maintain.

Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>JSON Fetch Practical</title>
</head>

<body>
  <h1>Student Details</h1>
  <div id="student"></div>
  <script>
    // Fetch JSON file
    fetch("students.json")
    // Convert response into JSON
    .then(response => response.json())
    // Get JSON data
    .then(students => {

      let output = "";
      // Display each student
      students.forEach(student => {
        output += `
        <h2>${student.name}</h2>
        <p>Age: ${student.age}</p>
        <p>Course: ${student.course}</p>
        <p>Marks: ${student.marks}</p>
        <hr>
        `;
      });

      // Display on webpage
      document.getElementById("student").innerHTML = output;
    })

    // Error handling
    .catch(error => {
      console.log("Error:", error);
    });
  </script>
</body>
</html>
```

students.JSON

```
[
  {
    "name": "Rahul",
    "age": 20,
    "course": "CSE",
    "marks": 85
  },
  {
    "name": "Priya",
    "age": 19,
    "course": "IT",
    "marks": 90
  },
  {
    "name": "Amit",
    "age": 21,
    "course": "CSE",
    "marks": 78
  }
]
```

Output:**Student Details****Rahul**

Age: 20

Course: CSE

Marks: 85

Priya

Age: 19

Course: IT

Marks: 90

Amit

Age: 21

Course: CSE

Marks: 78